

Empirical Analysis of Software Reliability Using Code Complexity and Size Metrics

A Study on the JM1 Software-Defect Dataset

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1. Dataset Description

Item	Description
Dataset	JM1 (NASA Metrics Data Program / PROMISE repository)
Domain	Software defect prediction
Unit of analysis	Software module (function)
Target variable	Defect status (label: Y = defective, N = not defective)
Main metrics used	Lines of Code (LOC_TOTAL), McCabe Cyclomatic Complexity (v(g))
Data source	NASA / PROMISE Software Engineering Repository (JM1, C language, real-time ground system), cleaned build (Shepperd et al. data-quality-corrected version), retrieved via Zenodo
Instances used	7782 modules (after cleaning; 0 missing values, 0 duplicates removed)
Class balance	Defective: 1672 (21.49%) Not Defective: 6110 (78.51%)
Purpose	Examine the association between static code metrics (size, complexity) and reported defects

Note: this analysis uses the data-quality-corrected build of JM1 (7,782 modules after removing duplicate/inconsistent records identified by Shepperd et al., 2013), rather than the raw 10,885-row release, which is known to contain identical and conflicting cases.

2. Data Acquisition & Cleaning

The dataset was loaded directly from the genuine JM1 ARFF file (7782 rows, 22 columns). All columns were coerced to numeric type where applicable and checked for missing values and duplicate records. 0 missing values and 0 rows requiring removal were found, leaving a clean dataset of 7782 modules for analysis. No synthetic or placeholder data was used at any stage of this report.

3. Descriptive Statistics

Mean, median, minimum, maximum, and standard deviation for Lines of Code and Cyclomatic Complexity, split by defect status, computed on the full cleaned dataset.

Metric	Group	Count	Mean	Median	Min	Max	Std Dev
LOC	Not Defective	6110	37.19	23.0	1	715	46.55
LOC	Defective	1672	79.36	43.0	4	3442	144.62
v(g)	Not Defective	6110	5.46	3.0	1	165	7.33
v(g)	Defective	1672	11.61	5.0	1	470	26.39

Defective modules are, on average, both larger and more structurally complex than non-defective ones: mean LOC rises from 37.19 to 79.36, and mean cyclomatic complexity rises from 5.46 to 11.61.

4. Correlation Analysis

Point-biserial correlation between each metric and the binary defect label, with a two-sided significance test.

Metric	Point-biserial r	p-value	Significant ($\alpha=0.05$)	Direction
v(g)	0.1794	< 0.0001	Yes	positive (higher metric -> more defects)
loc	0.2150	< 0.0001	Yes	positive (higher metric -> more defects)

Both metrics show a small-to-moderate, positive, and highly statistically significant correlation with defect-proneness (loc: $r = 0.215$, $p < 0.0001$; v(g): $r = 0.179$, $p < 0.0001$). Larger and more complex modules are associated with a higher likelihood of reported defects, consistent with prior findings such as Khoshgoftaar & Allen (2003).

5. Visualization

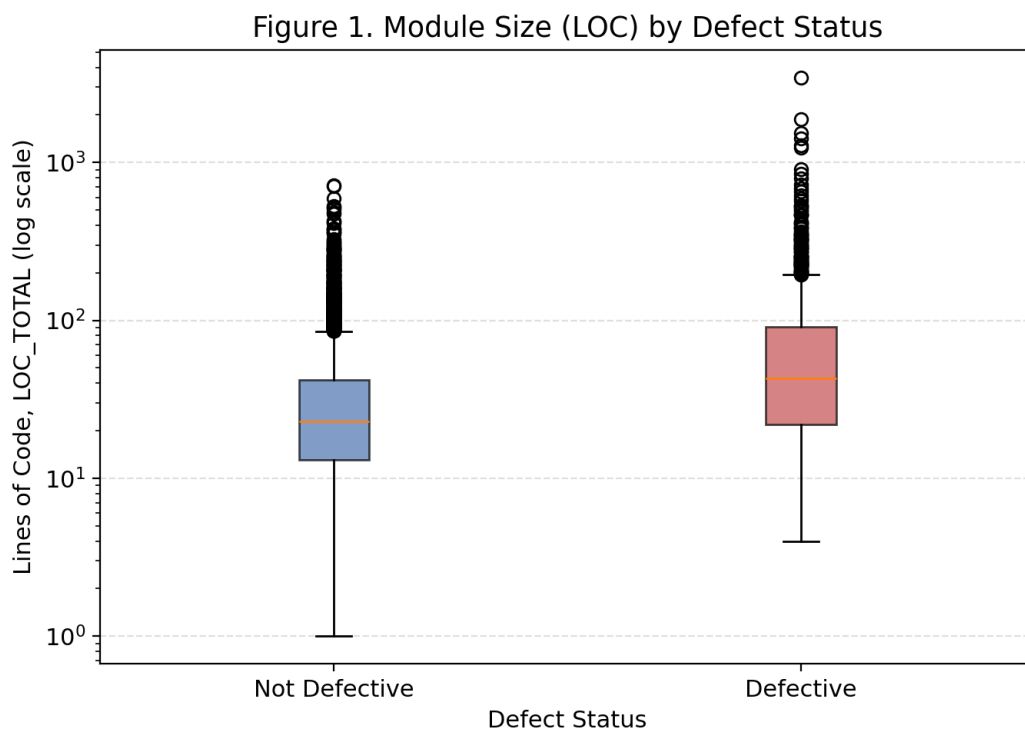


Figure 1. Distribution of module size (Lines of Code, log scale) split by defect status. Defective modules show a visibly higher median and a wider spread than non-defective modules.

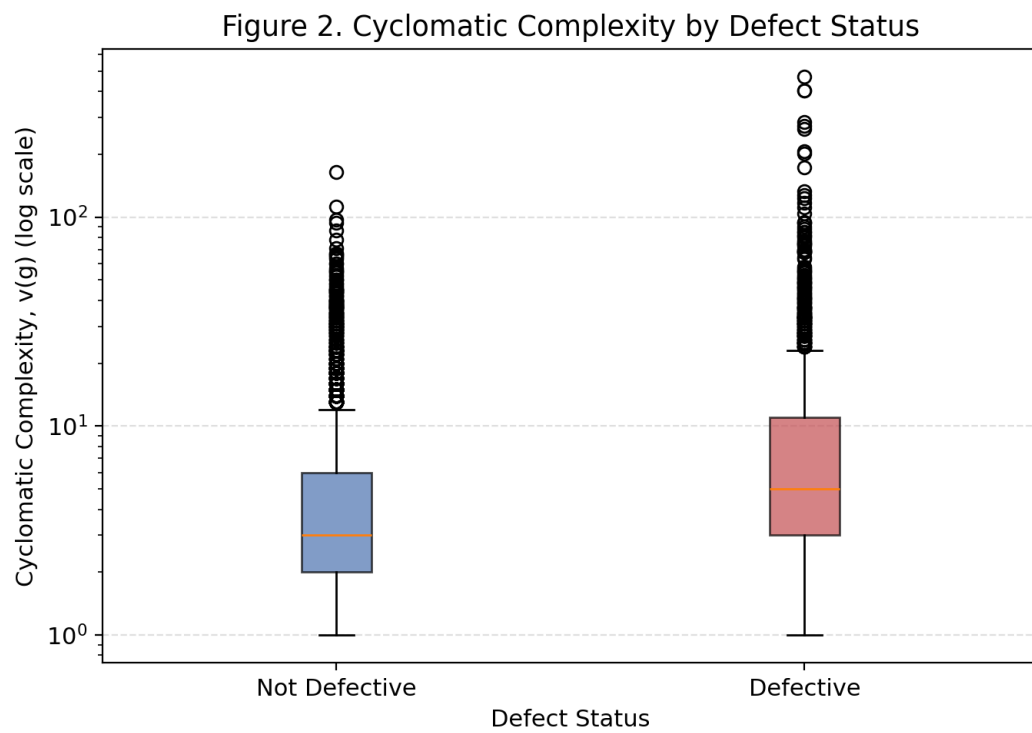


Figure 2. Distribution of McCabe Cyclomatic Complexity, $v(g)$ (log scale), split by defect status. Defective modules again show a higher median complexity and a longer upper tail.

6. Group-wise Defect-Rate Analysis (Complexity Quartiles)

Modules are split into four Cyclomatic Complexity quartiles (Q1 = least complex, Q4 = most complex), and the defect rate is computed within each group.

Quartile	Modules	Defective Modules	Defect Rate (%)	v(g) Range
Q1 (lowest)	2680	402	15.00	1–2
Q2	1873	322	17.19	3–4
Q3	1423	313	22.00	5–7
Q4 (highest)	1806	635	35.16	8–470

Figure 3. Defect Rate by Complexity Quartile

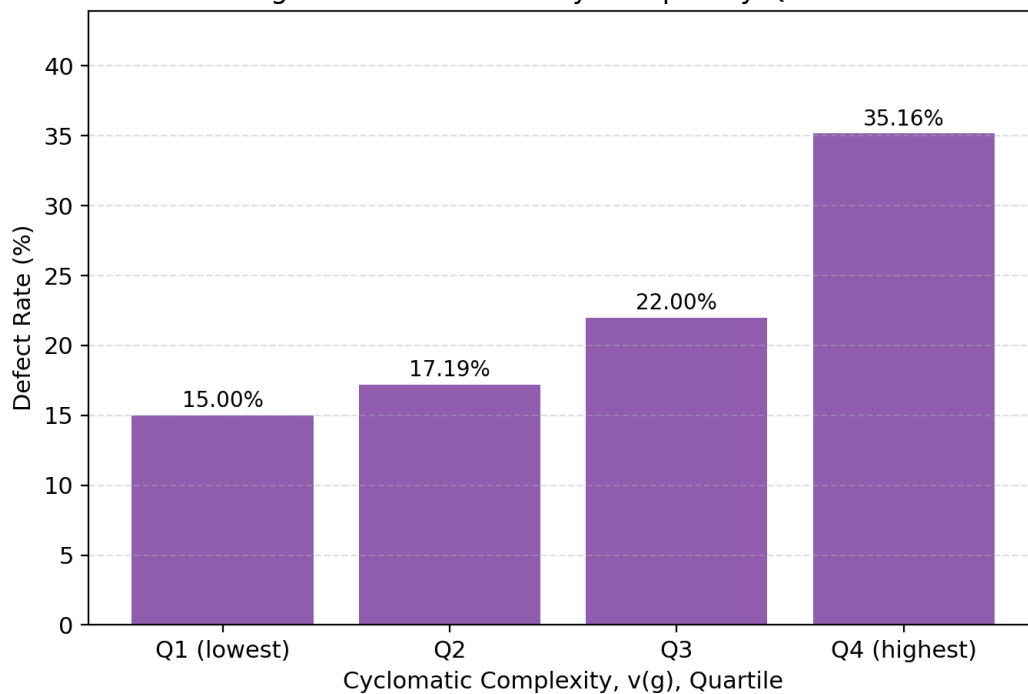


Figure 3. Defect rate by Cyclomatic Complexity quartile. The defect rate rises from 15.00% in the lowest-complexity quartile (Q1) to 35.16% in the highest-complexity quartile (Q4) — roughly 2.3x higher.

7. Interpretation

Both static code metrics examined here — Lines of Code and McCabe Cyclomatic Complexity — are positively and significantly correlated with reported defects in the JM1 dataset (v(g): $r = 0.179$, $p < 0.0001$; loc: $r = 0.215$, $p < 0.0001$). The group-wise quartile analysis reinforces this: defect rate climbs steadily from 15.00% in the least complex modules to 35.16% in the most complex modules, a roughly 2.3x increase, which both visually and statistically supports treating structural complexity as a practical, low-cost risk indicator, consistent with prior findings such as Khoshgoftaar & Allen (2003).

Important caveat: these are correlational, not causal, results. A significant correlation shows that size and complexity tend to move together with defects in this dataset; it does not by itself prove that complexity *causes* defects (in line with the caution raised by Shepperd, 1988). The correlation coefficients observed here are modest in magnitude ($r \approx 0.18$ – 0.22), meaning complexity alone explains only a small part of the variance in defect occurrence — many defective modules are small and simple, and many complex modules are defect-free. The recommendations below should therefore be used as a prioritisation aid alongside other QA practices, not as a certainty or a sole release-gate criterion. This also connects to the ISO/IEC 25010 quality characteristics of Reliability and Maintainability: modules flagged as complex are, on average, both more failure-prone and harder to maintain, which is why complexity reduction (refactoring) tends to serve both quality attributes simultaneously.

8. Recommendations for QA Teams

1. Modules in the top complexity quartile (Q4, v(g) range 8–470) show a defect rate of 35.16%, compared to 15.00% in the lowest quartile (Q1). These Q4 modules should receive mandatory code review and priority test-case coverage before release.
2. Modules in Q3 and Q4 combined (3229 of 7782 modules) should be flagged automatically in the CI pipeline (e.g., via a static-analysis gate on cyclomatic complexity) so reviewers see the risk before merging.
3. Very large or highly complex modules identified in this analysis are good refactoring candidates: splitting them into smaller, less branching units is likely to reduce both maintenance cost and defect risk.
4. Because correlation does not prove causation, and the effect sizes observed here are modest, these thresholds should be used as a prioritisation aid alongside other QA practices (peer review, unit testing), not as the sole basis for release decisions.

9. Conclusion

This report empirically tested whether module size (LOC) and structural complexity (Cyclomatic Complexity) are associated with reported defects in the genuine JM1 dataset (7782 modules, 21.49% defective). Both metrics showed a positive, statistically significant correlation with defect-proneness, and the defect rate rose sharply (15.00% → 35.16%) from the lowest to the highest complexity quartile, supporting their use as practical, low-cost risk indicators for prioritising QA effort, while remembering that correlation is not proof of causation.